

Governance

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Summary

The implementation of CFM processes is patchy. The legal processes required for establishment of a CF are mostly complete across our sample, but later processes such as the registration certificate, forest-use permits and benefit-sharing agreement have not been completed for a substantial number of CFs. While most CFMGs have held elections, have term-limits for executive officers and hold meetings with reasonable frequency, a large proportion have not opened a bank account, do not have a receipt book and have not completed annual financial and technical reports. Most CFMGs are overdue elections and hence are not compliant with their Bylaws & Constitutions. This is worrying as, in some cases, it may reflect an attempt by CFMG executives to retain control for personal benefit. Business processes are even less well implemented. Many do not have a business plan, annual budgets or workplans. Many also do not have defined user groups or issue permits. Most CFMGs are only rarely visited by DFOs or other district forestry staff, but most are supported by conservation NGOs or carbon trading enterprises, who do visit more frequently. Interest in expanding the CF, an indicator of satisfaction with CFM, was highest among CFMG executives and lowest among community youth, a difference that was significant. Among individual interview respondents, NGO representatives indicated they would like to expand the CF significantly more often than traditional leaders (usually headmen or headwomen) and community members. For those CFMGs that recognised forest user groups, these user-groups harvested a very wide variety of resources, but honey and mushrooms were particularly frequently mentioned.

Introduction

In this chapter, we consider questions concerning the governance and management of CFMGs. As our questions pertain to the running of the CFMGs, our analysis focuses on the responses of the CFMG executives. However, the responses from other FGDs and from the individual interviews are included for comparison where appropriate.

CFMG legal processes

We asked CFMG executives what required legal processes had been completed, including the recognition letter, forest management agreement, forest management plan, community forest registration certificate, forest user-rights permit, CFMG bylaws & constitution, and benefit sharing agreement (Table 1).

```
cfmg |> filter(interview_type == "cfmg_executive") |>
  group_by(province) |>
  summarise(
    `CFMG recognition` = mean(
      cfm_governance_legal_documents_recognition_letter,
      na.rm = TRUE),
    `CF agreement` = mean(
      cfm_governance_legal_documents_community_forest_management_agreement,
      na.rm = TRUE),
    `Management plan` = mean(
      cfm_governance_legal_documents_forest_management_plan,
      na.rm = TRUE),
    `Registration certificate` = mean(
      cfm_governance_legal_documents_community_forest_registration_certificate,
      na.rm = TRUE),
    `User-rights permit` = mean(
      cfm_governance_legal_documents_forest_user_rights_license_permit,
      na.rm = TRUE),
    `Bylaws & constitution` = mean(
      cfm_governance_legal_documents_bylaws_and_constitution,
      na.rm = TRUE),
    `Benefit sharing agreement` = mean(
      cfm_governance_legal_documents_benefit_sharing_agreement,
      na.rm = TRUE)
  ) |>
  gt() |>
  fmt_percent(columns = -province, decimals = 0) |>
  tab_header(title = "CFMG legal processes",
             subtitle = "Values represent the % of CFMGs with each legal document by province") |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "center", columns = -province) |>
  cols_align(align = "left", columns = province) |>
  tab_style(
    style = cell_text(align = "center"),
    locations = cells_column_labels(columns = province)
```

```
) |>
sub_missing()
```

Table 1: CFMG completed legal processes by province

CFMG legal processes
Values represent the % of CFMGs with each legal document by province. n

province	CFMG recongnition	CF agreement	Management plan	Registration certificate	User-rights
Central	91%	82%	82%	55%	
Eastern	83%	75%	67%	8%	
Lusaka	100%	100%	100%	100%	100%
Muchinga	93%	93%	86%	57%	
North-Western	100%	100%	80%	60%	
Northern	100%	100%	91%	91%	
Southern	100%	100%	100%	50%	
Western	100%	83%	83%	67%	

Lusaka Province was the only province in which all processes had been completed across all CFMGs, while Eastern Province had the highest proportion of incomplete legal processes. Overall, completion of legal processes was quite patchy. Some processes, such as the Recognition letter, CF Agreement, Management Plan and Bylaws & Constitution, seem to have been completed for most CFMGs, but the Registration Certificate, User-rights Permit and Benefit-sharing Agreement were less often completed. This assessment is based on the responses of the CFMG executives, who should be well informed. Nevertheless, it may be that more processes have been completed, especially where the CFMG is being supported by external partners, such as NGOs or carbon trading agencies.

CFMG governance processes

Through the survey, we asked CFMG executives what governance processes were being implemented, such as holding CFMG elections, issuing permits and annual reporting (Table 2).

```
cfmg |> filter(interview_type == "cfmg_executive") |>
  mutate(
    cfmg_elections = case_match(
      cfmg_governance_cfmg_held_election,
      "yes" ~ 1,
      "no" ~ 0,
      .default = NA_integer_),
```

```

cfmg_term_limits = case_match(
  cfmg_governance_term_limits_exist,
  "yes" ~ 1,
  "no" ~ 0,
  .default = NA_integer_),
cfmg_HFO = case_match(
  cfmg_governance_yn_employ_honorary_forest_officers,
  "yes" ~ 1,
  "no" ~ 0,
  .default = NA_integer_),
cfmg_receipt_book = case_match(
  cfmg_governance_receipt_book_maintained,
  "yes" ~ 1,
  "no" ~ 0,
  .default = NA_integer_),
cfmg_bank_acc = case_match(
  cfmg_governance_bank_account,
  "yes" ~ 1,
  "no" ~ 0,
  .default = NA_integer_),
cfmg_financial_rep = case_match(
  cfmg_governance_financial_reports_to_community,
  "yes" ~ 1,
  "no" ~ 0,
  .default = NA_integer_),
cfmg_annual_rep = case_match(
  cfmg_governance_annual_report_to_community,
  "yes" ~ 1,
  "no" ~ 0,
  .default = NA_integer_),
cfmg_permits = case_match(
  cfmg_governance_issue_licenses_permits,
  "yes" ~ 1,
  "no" ~ 0,
  .default = NA_integer_)

) |>
group_by(province) |>
summarise(
  `CFMG elections` = mean(
    cfmg_elections,
    na.rm = TRUE),

```

```

`CFMG term-limits` = mean(
  cfm_term_limits,
  na.rm = TRUE),
`Receipt book` = mean(
  cfm_receipt_book,
  na.rm = TRUE),
`Bank acc` = mean(
  cfm_bank_acc,
  na.rm = TRUE),
`Permits issued` = mean(
  cfm_permits,
  na.rm = TRUE),
`Financial reporting` = mean(
  cfm_financial_rep,
  na.rm = TRUE),
`Annual reporting` = mean(
  cfm_annual_rep,
  na.rm = TRUE),
`HFOs appointed` = mean(
  cfm_HFO,
  na.rm = TRUE)
) |>
gt() |>
fmt_percent(columns = -province, decimals = 0) |>
tab_header(title = "CFMG governance processes",
  subtitle = "Values represent the % of CFMGs completing/conducting each gov
opt_align_table_header(align = "left") |>
cols_align(align = "center", columns = -province) |>
cols_align(align = "left", columns = province) |>
tab_style(
  style      = cell_text(align = "center"),
  locations = cells_column_labels(columns = province)
) |>
sub_missing()

```

Table 2: CFMG governance processes conducted by province

CFMG governance processes
Values represent the % of CFMGs completing/conducting each governance process
75 CFMGs.

province	CFMG elections	CFMG term-limits	Receipt book	Bank acc	Permits issued	Financial
Central	100%	55%	9%	0%	0%	
Eastern	83%	73%	75%	83%	17%	
Lusaka	100%	33%	33%	83%	0%	
Muchinga	100%	75%	55%	36%	33%	
North-Western	100%	88%	50%	10%	30%	
Northern	100%	73%	29%	80%	0%	
Southern	75%	25%	25%	0%	0%	
Western	100%	86%	0%	0%	0%	

While most CFMGs have held elections, have term limits for executive officers and employ honorary forest officers (HFOs), other governance processes were not as consistently implemented. Most CFMGs have not opened bank accounts, do not maintain a receipt book, or issue forest-use permits. Financial and annual reporting to community members was also apparently rarely implemented.

It appears that most CFMGs have not held elections since they were formed, up to five years ago (Figure 1).

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  mutate(
    election = case_match(
      cfmg_governance_last_cfmg_election_when,
      "Less than a month" ~ "<1 year ago",
      "less_than_1_year_ago" ~ "<1 year ago",
      "1_2_years_ago" ~ "1 to 2 years ago",
      "2 years" ~ "1 to 2 years ago",
      "2 years ago" ~ "1 to 2 years ago",
      "3_5_years_ago" ~ "3 to 5 years ago",
      "3 years" ~ "3 to 5 years ago",
      "3 years ago" ~ "3 to 5 years ago",
      "4 years ago" ~ "3 to 5 years ago",
      "more_than_5_years_ago" ~ ">5 years ago",
      .default = NA
    ),
    election = factor(election,
      levels = c(">5 years ago", "3 to 5 years ago", "1 to 2 years ago", "<1 year ago"))
  ) |>
  drop_na(election) |>
  count(election) |>
  mutate(prop = n / sum(n), group = "All CFMGs") |>
```

```

ggplot(aes(y = group, x = prop, fill = election)) +
  geom_col(width = 0.4, colour = "white", linewidth = 0.4) +
  geom_text(
    aes(label = ifelse(prop >= 0.06, n, "")),
    position = position_stack(vjust = 0.5),
    size = 3.5, fontface = "bold", colour = "white"
  ) +
  scale_x_continuous(
    labels = percent_format(accuracy = 1),
    expand = expansion(add = c(0, 0.02)),
    name = "Proportion of CFMGs"
  ) +
  scale_fill_manual(
    values = c(
      "<1 year ago" = "#2A9D8F",
      "1 to 2 years ago" = "#76B7A0",
      "3 to 5 years ago" = "#E07B54",
      ">5 years ago" = "#C0392B"
    ),
    guide = guide_legend(reverse = TRUE)
  ) +
  labs(
    y = NULL,
    fill = NULL
  ) +
  scale_y_discrete(expand = expansion(add = c(0.3, 0.3))) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank(),
    axis.text.x = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x = element_text(size = rel(1), colour = "grey40"),
    legend.position = "bottom",
    legend.text = element_text(size = rel(1)),
    plot.margin = margin(2, 25, 10, 10)
  )

```

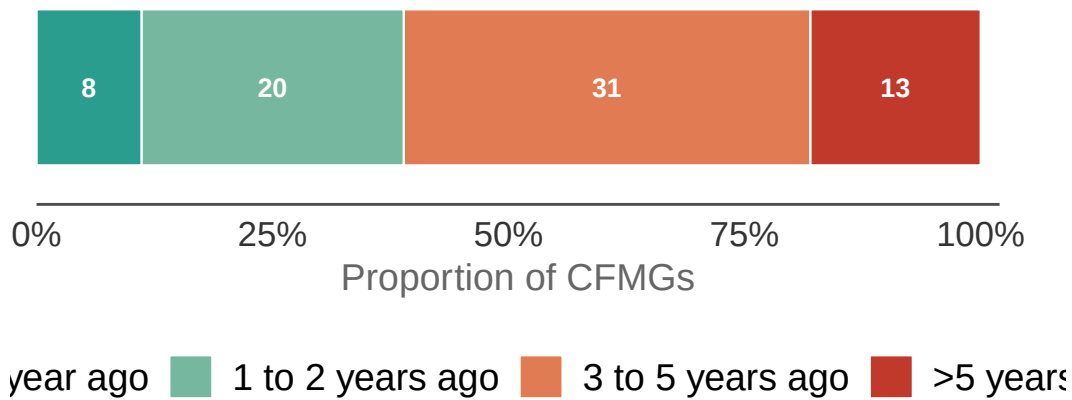


Figure 1: Number of years since the last CFMG executive election. The length of each bar segment represents the proportion of CFMGs in that category. The numbers superimposed on the bars represent the actual number of CFMGs in each category.

Forty-three out of 75 CFMGs have not held elections for over 3 yrs, although term limits for executive officers are typically 2-3 years (Figure 2). This indicates that most CFMGs are not complying with their Bylaws & Constitutions.

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  mutate(
    tenure = case_match(
      cfmg_governance_cfmg_postions_tenure,
      "2_years" ~ "2 years",
      "3_years" ~ "3 years",
      "5 time" ~ "5 years",
      "5years" ~ "5 years",
      "6 years" ~ ">5 years",
      "7 years" ~ ">5 years",
      "dont_know" ~ NA,
      .default = NA
    ),
    tenure = factor(tenure, levels = c(">5 years", "5 years", "3 years", "2 years"))
  ) |>
  drop_na(tenure) |>
  count(tenure) |>
  mutate(prop = n / sum(n), group = "All CFMGs") |>
  ggplot(aes(y = group, x = prop, fill = tenure)) +
```



```

geom_col(width = 0.4, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.06, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(add = c(0, 0.02)),
  name = "Proportion of CFMGs"
) +
scale_fill_manual(
  values = c(
    "2 years" = "#2A9D8F",
    "3 years" = "#76B7A0",
    "5 years" = "#E07B54",
    ">5 years" = "#C0392B"
  ),
  guide = guide_legend(reverse = TRUE)
) +
labs(
  y = NULL,
  fill = NULL
) +
scale_y_discrete(expand = expansion(add = c(0.3, 0.3))) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y = element_blank(),
  axis.ticks.y = element_blank(),
  axis.text.x = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x = element_text(size = rel(1), colour = "grey40"),
  legend.position = "bottom",
  legend.text = element_text(size = rel(1)),
  plot.margin = margin(2, 25, 10, 10)
)

```

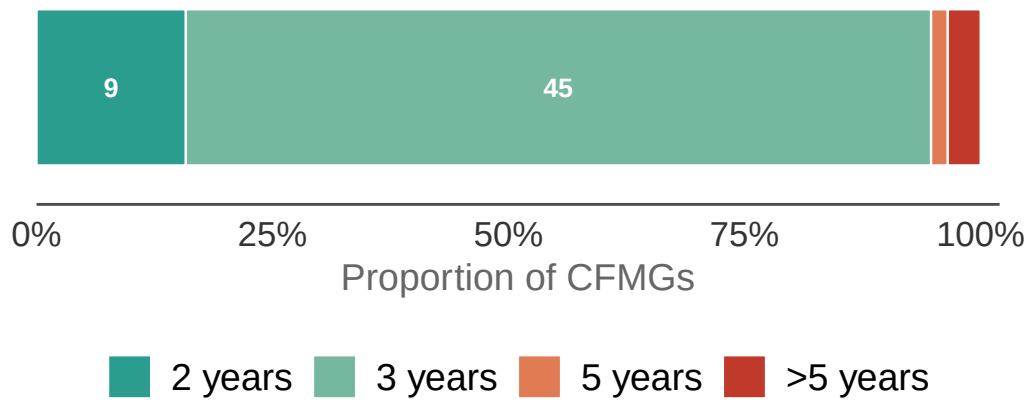


Figure 2: Tenure of CFMG executive positions. The length of each bar segment represents the proportion of CFMGs with that tenure length. The numbers superimposed on the bars represent the actual number of CFMGs in each category.

In contrast, most CFMGs do appear to be holding meetings regularly (Figure 3). However, there is a small proportion of CFMGs that holds meetings rarely or never.

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  mutate(
    meetings = case_match(
      cfmg_governance_cfmg_meetings,
      "Bi monthly" ~ ">4 yr-1",
      "monthly" ~ ">4 yr-1",
      "Quarterly" ~ "4 yr-1",
      "quarterly" ~ "4 yr-1",
      "yearly" ~ "1 yr-1",
      "only_when_needed_adhoc_meetings " ~ "rarely",
      "Due to lack of support the Executive does not meet often" ~ "rarely",
      "Quarterly but the executive has not had meetings in the last 2 years" ~ "never",
      "Quarterly but they don't meet" ~ "never",
      .default = NA
    ),
    meetings = factor(meetings,
      levels = c("never", "rarely", "1 yr-1", "4 yr-1", ">4 yr-1"))
  ) |>
  drop_na(meetings) |>
  count(meetings) |>
```

```

mutate(prop = n / sum(n), group = "All CFMGs") |>
ggplot(aes(y = group, x = prop, fill = meetings)) +
geom_col(width = 0.4, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.06, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(add = c(0, 0.02)),
  name = "Proportion of CFMGs"
) +
scale_fill_manual(
  values = c(
    ">4 yr-1" = "#2A9D8F",
    "4 yr-1" = "#76B7A0",
    "1 yr-1" = "#F0C040",
    "rarely" = "#E07B54",
    "never" = "#C0392B"
  ),
  labels = c(
    ">4 yr-1" = expression(">4 yr"^-1),
    "4 yr-1" = expression("4 yr"^-1),
    "1 yr-1" = expression("1 yr"^-1),
    "rarely" = "rarely",
    "never" = "never"
  ),
  guide = guide_legend(reverse = TRUE)
) +
labs(
  y = NULL,
  fill = NULL
) +
scale_y_discrete(expand = expansion(add = c(0.3, 0.3))) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y = element_blank(),

```

```

axis.ticks.y      = element_blank(),
axis.text.x       = element_text(size = rel(1.14), colour = "grey20"),
axis.title.x      = element_text(size = rel(1), colour = "grey40"),
legend.position   = "bottom",
legend.text       = element_text(size = rel(1)),
plot.margin       = margin(2, 25, 10, 10)
)

```

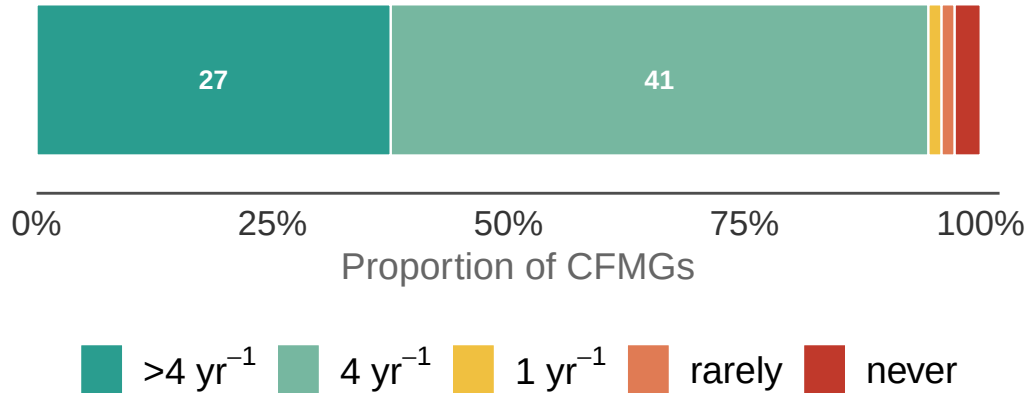


Figure 3: Number of CFMG meetings per year. The length of each bar segment represents the proportion of CFMGs with the specified meeting frequency. The numbers superimposed on the bars represent the actual number of CFMGs in each category.

CFMG business processes

As with the legal processes, we asked CFMG executives about the state of their business processes. Specifically, we asked if they had developed annual workplans and budgets, and whether these plans were being implemented. We also asked if they had developed a business plan and if any forest user groups had been established (Table 3).

```

cfmg |> filter(interview_type == "cfmg_executive") |>
  mutate(
    cfmg_workplan = case_match(
      cfmg_governance_cfmg_workplan,
      "yes" ~ 1,
      "no" ~ 0,
      "don't_know" ~ NA),
    cfmg_budget = case_match(

```

```

        cfm_governance_cfm_budget,
        "yes" ~ 1,
        "no" ~ 0,
        "don't know" ~ NA),
    cfm_business_plan = case_match(
        cfm_governance_cfma_business_plan,
        "yes" ~ 1,
        "no" ~ 0,
        "don't know" ~ NA),
    cfm_user_groups = case_match(
        cfm_governance_any_user_groups,
        "yes" ~ 1,
        "no" ~ 0,
        "don't know" ~ NA)
) |>
group_by(province) |>
summarise(
    `CFMG workplan` = mean(
        cfm_workplan,
        na.rm = TRUE),
    `CFMG budget` = mean(
        cfm_budget,
        na.rm = TRUE),
    `Implementing plan` = mean(
        cfm_governance_implementing_plans_cfm_executive,
        na.rm = TRUE),
    `Business plan` = mean(
        cfm_business_plan,
        na.rm = TRUE),
    `User groups` = mean(
        cfm_user_groups,
        na.rm = TRUE)
) |>
mutate(across(where(is.numeric), ~ifelse(is.nan(.x), NA_real_, .x))) |>
gt() |>
fmt_percent(columns = -province, decimals = 0) |>
tab_header(title = "Business processes in the management of CFMGs",
            subtitle = "Values represent the % of CFMGs with each business process in
opt_align_table_header(align = "left") |>
cols_align(align = "center", columns = -province) |>
cols_align(align = "left", columns = province) |>
tab_style(

```

```

    style      = cell_text(align = "center"),
    locations = cells_column_labels(columns = province)
  ) |>
  sub_missing()

```

Table 3: CFMG completed business process by province.

Business processes in the management of CFMGs

Values represent the % of CFMGs with each business process in place by province. n = 75 CFMGs.

province	CFMG workplan	CFMG budget	Implementing plan	Business plan	User groups
Central	62%	27%	100%	27%	82%
Eastern	—	83%	92%	58%	50%
Lusaka	67%	67%	67%	17%	33%
Muchinga	100%	43%	100%	71%	86%
North-Western	30%	20%	60%	20%	60%
Northern	—	73%	100%	45%	73%
Southern	100%	75%	100%	0%	50%
Western	67%	0%	100%	33%	100%

As is evident from the table, these business management processes were even more erratically completed than the legal and governance processes. As substantial proportion of CFMGs did not have workplans, budgets or a business plan. More surprising, many did not have any recognised user groups, which is critical to managing forest resources.

External support

All CFMGs are supported by FD, which has a statutory duty to implement the CFM policy and regulations. Nonetheless, the frequency of visits by DFOs or other district forestry staff varied widely and in many cases they rarely or never visited.

```

cfmg |>
  filter(interview_type == "cfmg_executive") |>
  mutate(
    dfo = case_match(dfo_visits,
      "never" ~ "Never",
      "rarely" ~ "Rarely",
      "occasionally" ~ "Occasionally",
      "often" ~ "Often",
      "very_often" ~ "Very often",

```

```

      .default      = NA
    ),
    dfo = factor(dfo, levels = c("Very often", "Often", "Occasionally", "Rarely", "Never"))
  ) |>
drop_na(dfo) |>
count(province, dfo) |>
group_by(province) |>
mutate(
  prop      = n / sum(n),
  mean_score = sum(as.numeric(dfo) * prop)
) |>
ungroup() |>
ggplot(aes(y = reorder(province, mean_score), x = prop, fill = dfo)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.08, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of CFMGs"
) +
scale_fill_manual(
  values = c(
    "Never"      = "#C0392B",
    "Rarely"     = "#E07B54",
    "Occasionally" = "#F0C040",
    "Often"      = "#76B7A0",
    "Very often" = "#2A9D8F"
  )
) +
  guides(fill = guide_legend(reverse = TRUE)) +
labs(
  y      = NULL,
  fill = NULL
) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),

```

```

panel.grid.major.x = element_blank(),
axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
axis.title.x       = element_text(size = rel(1), colour = "grey40"),
legend.position    = "bottom",
legend.text        = element_text(size = rel(1)),
plot.margin        = margin(10, 25, 10, 10)
)

```

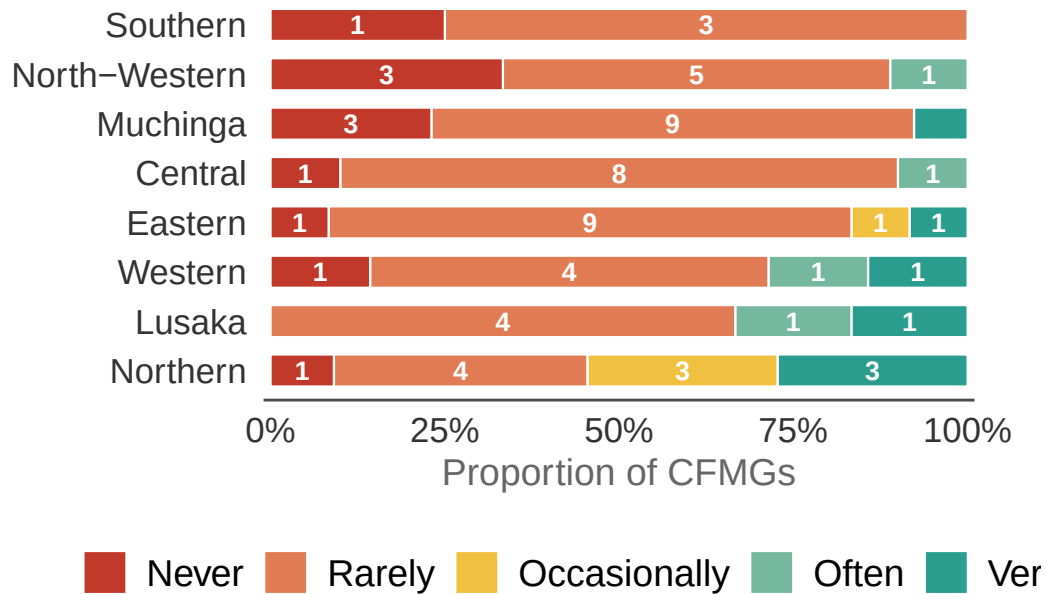


Figure 4: Frequency of visits by District Forest Officers (DFO) or other district forestry staff by province. The length of the bars represents the proportion of CFMGs with a specified frequency of visits. The numbers superimposed on the bars represent the actual number of CFMGs in each category.

Many CFMGs also operate in partnership with an NGO, such as a conservation organisation, or a private enterprise, such as a carbon trader (Figure 5).

```

cfmg |>
  filter(interview_type == "cfmg_executive") |>
  mutate(
    response = substr(cfmg_governance_yn_institutions_support_cfmg, 1, 1),

```



```

response = case_match(
  response,
  "y" ~ "Yes",
  "n" ~ "No"
)
) |>
filter(!is.na(response)) |>
count(province, response) |>
mutate(
  n_signed = if_else(response == "No", -n, n),
  response = factor(response, levels = c("No", "Yes"))
) |>
ggplot(aes(y = reorder(province, n_signed), x = n_signed, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
geom_text(
  aes(label = n,
      hjust = if_else(response == "No", 1.4, -0.4)),
  size = 3.5, fontface = "bold", colour = "grey25"
) +
scale_x_continuous(
  labels = ~abs(.x),
  expand = expansion(mult = 0.18),
  name = "Number of CFMGs"
) +
scale_fill_manual(
  values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
  guide = guide_legend(reverse = FALSE)
) +
labs(
  y = NULL,
  fill = NULL
) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x = element_text(size = rel(1), colour = "grey40"),

```

```

legend.position = "bottom",
legend.text     = element_text(size = rel(1)),
plot.margin    = margin(10, 25, 10, 10)
)

```

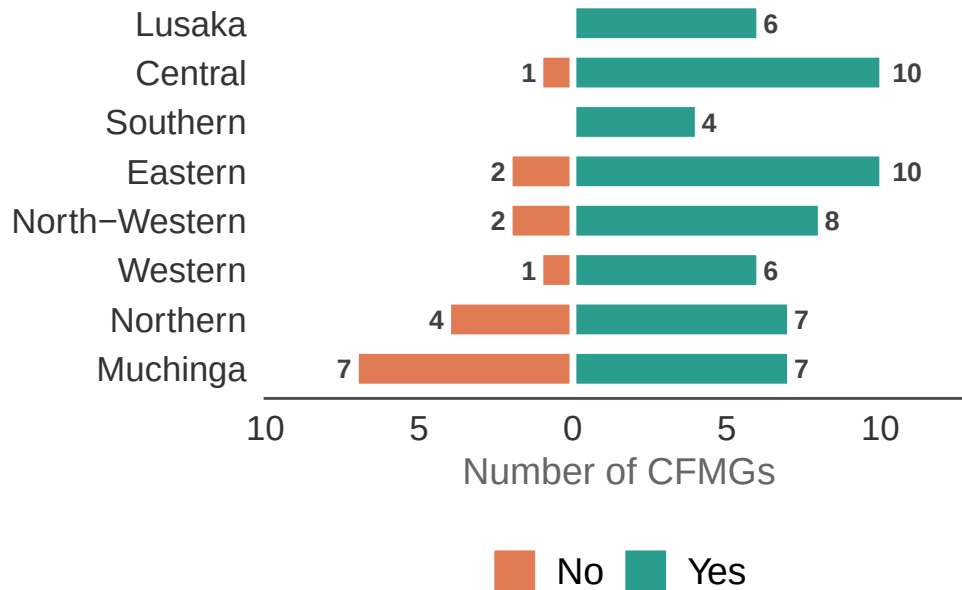


Figure 5: CFMGs partnered with an external organisation, such as a conservation NGO or carbon enterprise, by province. The length of the bars represents the number of CFMGs either partnered (Yes) or not partnered (No) with an external organisation. The numbers on the bars represent the actual number of CFMGs in each category.

The table below (Table 4) lists the partners organisations and the number of CFMGs they support. It is important to note that this list does not cover all organisations supporting CFMGs or all the CFMGs that are supported by a named organisation, but only those in our sample of 75 CFMGs. The FD is named as a partner in situations where the district FD office was responsible for setting up the CFMG and other partners are not involved.

```

cfmg |> filter(interview_type == "cfmg_executive") |>
  mutate(
    `CFMG partner` = cfmg_governance_support_institutions_supportinstitutions,
    `CFMG partner` = case_match(
      `CFMG partner`,
      "EP-JSLP" ~ "Forestry Department",

```

```

        "Community Climate Solutions Zambia" ~ "Community Climate Solutions",
        .default = `CFMG partner`
    )
  ) |>
count(`CFMG partner`) |>
gt() |>
tab_header(title = "Partner organisations",
           subtitle = "n = number of CFMGs partnered with each organisation.") |>
opt_align_table_header(align = "left") |>
cols_align(align = "center", columns = -`CFMG partner`) |>
cols_align(align = "left", columns = `CFMG partner`) |>
tab_style(
  style      = cell_text(align = "center"),
  locations = cells_column_labels(columns = `CFMG partner`)
) |>
sub_missing()

```

Table 4: Organisations partnering with CFMGs

Partner organisations
n = number of CFMGs partnered with each organisation.

CFMG partner	n
BioCarbon Patners	9
COMACO	11
Community Climate Solutions	8
DNPW	3
Forestry Department	4
GIZ	1
Kabompo Carbon Community	3
Kasanka Trust Ltd	2
South Luangwa Conservation	1
TRALARD	2
WWF	6
WeForest	4
West Lunga Conservation Project	3
—	18

The figure below (Figure 6) indicates the frequency of visits to CFMGs by partner organisations. The respondents did not state a specified number of visits per year, but from the context

we assume the frequently indicates monthly or every other month, occasionally indicates 1-4 times a year, while never means that they had not visited in over a year.

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  mutate(
    `Support freq` = cfmg_governance_support_institutions_quests_supportinstitutionsdetails_1,
    `Support freq` = case_match(`Support freq`,
                                "never_0_times" ~ "Never",
                                "twice_2_times" ~ "Occasionally",
                                "thrice_3_times" ~ "Occasionally",
                                "once_1_time" ~ "Occasionally",
                                "more_than_5_times" ~ "Occasionally",
                                "more_than_3_times" ~ "Occasionally",
                                "More than 3 times" ~ "Occasionally",
                                "More 3" ~ "Occasionally",
                                "Dependent on their activities" ~ "Occasionally",
                                "8_times" ~ "Occasionally",
                                "4_times" ~ "Occasionally",
                                "12_time (at_least_every_month)" ~ "Frequently",
                                "12_times (every_month)" ~ "Frequently",
                                "36_times" ~ "Frequently",
                                "COMACO 15 to 20 times per year" ~ "Frequently",
                                "Very often" ~ "Frequently",
                                "Very often/regularly. They have a Field Officer living in the community" ~ "Frequently",
                                "Very often" ~ "Frequently",
                                "more_than_12_times" ~ "Frequently",
                                "several/often_times" ~ "Frequently",
                                "very_often" ~ "Frequently",
                                "very_often (60_meetings_per_quater)" ~ "Frequently",
                                "very_often (they_have_a_lot_of_activities_with_the_community)" ~ "Frequently",
                                "very_often (they_have_an_officer_within_the_community)" ~ "Frequently",
                                .default = NA),
    `Support freq` = factor(`Support freq`, levels = c("Frequently", "Occasionally", "Never"), ordered = TRUE),
    `CFMG partner` = cfmg_governance_support_institutions_quests_supportinstitutionsdetails_1,
    `CFMG partner` = case_match(
      `CFMG partner`,
      "EP-JSLP" ~ "Forestry Department",
      "Community Climate Solutions Zambia" ~ "Community Climate Solutions",
      .default = `CFMG partner`
    )
  ) |>
  drop_na(`Support freq`) |>
```

```

    drop_na(`CFMG partner`) |>
count(`CFMG partner`, `Support freq`) |>
group_by(`CFMG partner`) |>
mutate(
  prop      = n / sum(n),
  mean_score = sum(as.numeric(`Support freq`) * prop)
) |>
ungroup() |>
ggplot(aes(y = reorder(`CFMG partner`, mean_score), x = prop, fill = `Support freq`)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.08, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of CFMGs"
) +
scale_fill_manual(
  values = c(
    "Never"      = "#C0392B",
    "Occasionally" = "#F0C040",
    "Frequently"  = "#2A9D8F"
  )
) +
  guides(fill = guide_legend(reverse = TRUE)) +
labs(
  y      = NULL,
  fill   = NULL
) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",

```

```
legend.text      = element_text(size = rel(1)),  
plot.margin      = margin(10, 25, 10, 10)  
)
```

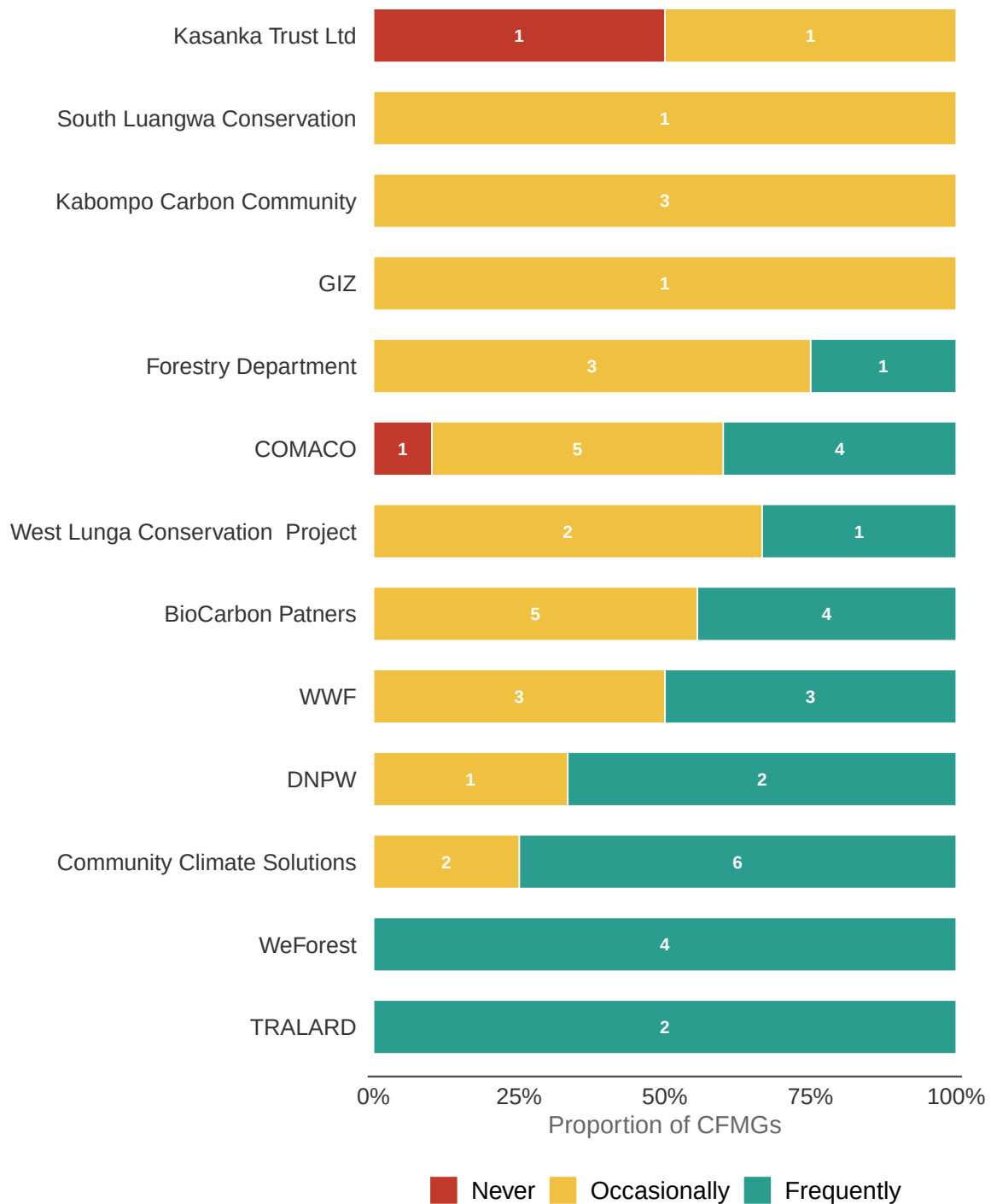


Figure 6: Frequency of visits by the supporting organisation. The length of the bars represents the proportion of CFMGs with a specified frequency of visits. The numbers superimposed on the bars represent the actual number of CFMGs in each category.

Interest in expanding the community forest

Next we asked interviewees whether they were interested in expanding the area of their CF. A “yes” answer can be taken as an endorsement of CFM. On the other hand, a negative response might reflect dissatisfaction with the CFMG management or the benefits CFM has brought. However, a negative response could also reflect other reasons, such as a lack of sufficient additional forest area in the community. We compare the responses from different FGD groups.

```
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    expand_cfma = substr(cfmg_governance_interest_expanding_cfma, 1, 1),
    expand_cfma = case_match(
      expand_cfma,
      "y" ~ "Yes",
      "n" ~ "No",
      "d" ~ "Don't know"
    ),
    expand_cfma = factor(expand_cfma, levels = c("Yes", "Don't know", "No")),
    interview_type = case_match(
      interview_type,
      "cfmg_executive" ~ "Executive",
      "cfmg_female" ~ "Female",
      "cfmg_male" ~ "Male",
      "cfmg_youth" ~ "Youth"
    )
  ) |>
  filter(!is.na(expand_cfma)) |>
  count(interview_type, expand_cfma) |>
  group_by(interview_type) |>
  mutate(
    prop = n / sum(n),
    mean_score = sum(as.numeric(expand_cfma) * prop)
  ) |>
  ungroup() |>
  ggplot(aes(y = reorder(interview_type, -mean_score), x = prop, fill = expand_cfma)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_text(
    aes(label = ifelse(prop >= 0.05, n, "")),
    position = position_stack(vjust = 0.5),
    size = 3.5, fontface = "bold", colour = "white"
  ) +
```



```

scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c(
    "No"           = "#C0392B",
    "Don't know"   = "#F0C040",
    "Yes"          = "#2A9D8F"
  ),
  guide = guide_legend(reverse = TRUE)
) +
labs(
  y      = NULL,
  fill   = NULL
) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position     = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin         = margin(10, 25, 10, 10)
)

```

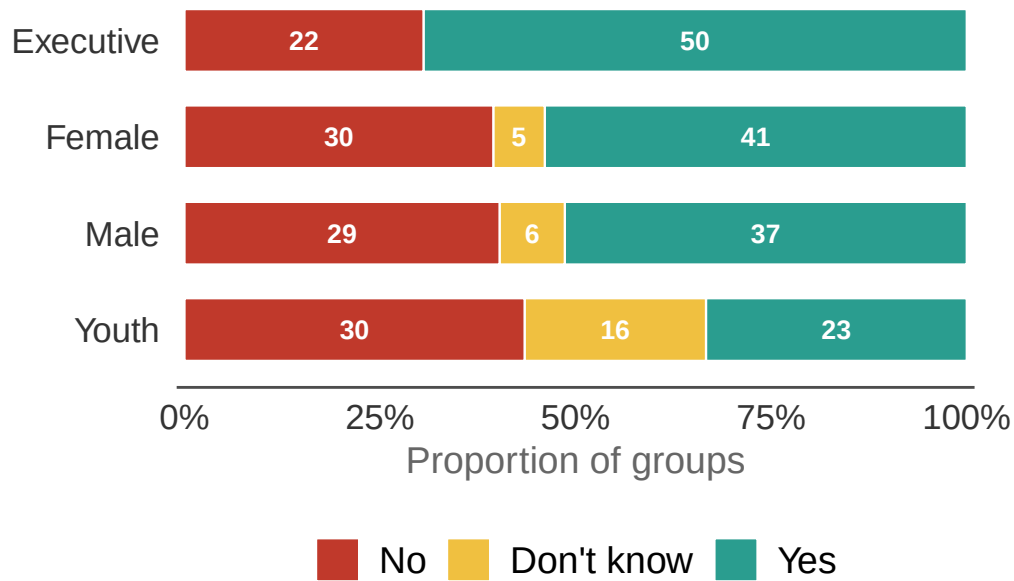


Figure 7: Interest in expanding the community forest by FGD type. The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

As can be seen in Figure 7, the CFMG executives were enthusiastic to expand the CF in most cases. However, among the community this number dropped to 41 for female groups, 37 for male groups and to just 23 for youth groups.

The difference in attitudes between CFMG executive and youth FGD groups was significant, but the pairwise comparisons among other FGD groups were not significant. These results might suggest that CFM is not benefiting youth to the same degree as other groups, or perhaps that they bear higher opportunity costs. Our analyses in other sections will help elucidate the reasons.

i Interest in expanding community forest - FGDs - Model methods

To examine the differences among FGD types, we employed a binomial generalised linear mixed model (GLMM) (i.e. yes = want to expand, no = do not want to expand), with random intercepts for province and CFMG nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```

test.data <- cfmg |> filter(interview_type != "individual_interview") |>
  mutate(
    expand_cfma = substr(
      cfmg_governance_interest_expanding_cfma, 1,1),
    expand_cfma = case_match(
      expand_cfma,
      "y" ~ "Yes",
      "n" ~ "No",
      "d" ~ "Don't know"
    ),
    expand_cfma = factor(expand_cfma,
      levels = c("Yes", "No", "Don't know")
    )
  ) |>
  mutate(
    expand_cfma = if_else(expand_cfma == "Don't know", NA, expand_cfma)
  )

M1 <- glmer(expand_cfma ~
  interview_type +
  gov_index +
  (1 | province/cfmg_name),
  family = binomial(link = "logit"),
  data = test.data
)

emm.M1 <- emmeans(M1, specs = ~ interview_type)
pairwise.M1 <- pairs(emm.M1, adjust = "Tukey")

```

```

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: expand_cfma ~ interview_type + gov_index + (1 | province/cfmg_name)
Data: test.data

```

AIC	BIC	logLik	deviance	df.resid
328.2	352.8	-157.1	314.2	240

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.6869	-0.7487	-0.4893	0.9195	1.9399

Random effects:

Groups	Name	Variance	Std.Dev.
cfmg_name:province	(Intercept)	4.446e-01	6.668e-01
province	(Intercept)	1.401e-09	3.743e-05

Number of obs: 247, groups: cfmg_name:province, 70; province, 8

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.10474	0.43893	0.239	0.81140
interview_typecfmg_female	0.43359	0.38663	1.121	0.26210
interview_typecfmg_male	0.71362	0.39391	1.812	0.07004 .
interview_typecfmg_youth	1.25027	0.41876	2.986	0.00283 **
gov_index	-0.22519	0.07918	-2.844	0.00445 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	intrvw_typcfmg_f	intrvw_typcfmg_m	intrvw_typcfmg_y
intrvw_typcfmg_f	-0.439			
intrvw_typcfmg_m	-0.400	0.496		
intrvw_typcfmg_y	-0.346	0.472	0.466	
gov_index	-0.752	-0.012	-0.053	-0.104

optimizer (Nelder-Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

contrast	estimate	SE	df	z.ratio	p.value
cfmg_executive - cfmg_female	-0.434	0.387	Inf	-1.121	0.6763
cfmg_executive - cfmg_male	-0.714	0.394	Inf	-1.812	0.2677
cfmg_executive - cfmg_youth	-1.250	0.419	Inf	-2.986	0.0150
cfmg_female - cfmg_male	-0.280	0.392	Inf	-0.714	0.8915
cfmg_female - cfmg_youth	-0.817	0.415	Inf	-1.969	0.1997
cfmg_male - cfmg_youth	-0.537	0.420	Inf	-1.277	0.5778

Results are given on the log odds ratio (not the response) scale.

P value adjustment: tukey method for comparing a family of 4 estimates

We also compared the responses from the individual interviewees (Table 5). The traditional leaders and community members more often reported that they did not want to expand the CF than other groups. However, the only pairwise comparisons that were significant were NGO representative with traditional leader and NGO representative with community member.

```

cfmg |> filter(interview_type == "individual_interview") |>
  mutate(
    expand_cfma = substr(cfmg_governance_interest_expanding_cfma, 1, 1),
    expand_cfma = case_match(expand_cfma, "y"~"Yes", "n"~"No", "d"~"Don't know"),
    expand_cfma = factor(expand_cfma, levels = c("Yes", "No", "Don't know")),
    interviewee_role = case_match(
      as.character(interviewee_role),
      "cfmg_exec_member" ~ "CFMG executive member",
      "cfmg_member"      ~ "Community member",
      "trad_leader"      ~ "Traditional leader",
      "dfo"              ~ "District Forestry Officer",
      "NGO_rep"          ~ "NGO representative"
    ),
    interviewee_role = factor(interviewee_role, levels = c(
      "CFMG executive member", "Community member", "Traditional leader",
      "District Forestry Officer", "NGO representative"
    ))
  ) |>
  filter(!is.na(expand_cfma)) |>
  count(interviewee_role, expand_cfma, name = "obs") |>
  group_by(interviewee_role) |>
  mutate(n = sum(obs), prop = obs / n) |>
  ungroup() |>
  select(interviewee_role, n, expand_cfma, prop) |>
  pivot_wider(names_from = expand_cfma, values_from = prop, values_fill = 0) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(interviewee_role = "Role") |>
  fmt_percent(columns = c(Yes, No, `Don't know`), decimals = 0) |>
  tab_header(title = "Interest in expanding the community forest area by interviewee role",
    subtitle = "Values represent the % of respondents in each response category. n = ") |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "center", columns = -interviewee_role) |>
  cols_align(align = "left", columns = interviewee_role) |>
  tab_style(
    style      = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()

```

Table 5: Interest in expanding community forest by FDG type.

Interest in expanding the community forest area by interviewee role

Values represent the % of respondents in each response category. n = the total number of interview responses.

Role	n	Yes	No	Don't know
CFMG executive member	62	66%	27%	6%
Community member	30	37%	53%	10%
Traditional leader	75	52%	43%	5%
District Forestry Officer	56	64%	25%	11%
NGO representative	32	81%	19%	0%

i Interest in expanding community forest - Individual interviewees - Model methods

To examine differences among individual interviewees in their stated interest in expanding their community forest, we employed a binomial GLMM (i.e. yes = want to expand, no = do not want to expand) and included interviewee role, interviewee gender and interviewee age as factors. We specified random intercepts for province and cfmg nested within province. To examine the pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |> filter(interview_type == "individual_interview") |>
  mutate(
    expand_cfma = substr(
      cfmg_governance_interest_expanding_cfma, 1,1),
    expand_cfma = case_match(
      expand_cfma,
      "y" ~ "Yes",
      "n" ~ "No",
      "d" ~ "Don't know"
    ),
    expand_cfma = factor(expand_cfma,
      levels = c("Yes", "No", "Don't know")
    )
  ) |>
  mutate(
    expand_cfma = if_else(expand_cfma == "Don't know", NA, expand_cfma),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age = demo_information_interviewee_age
  )
```

```

M2 <- glmer(expand_cfma ~
              interviewee_role +
              interviewee_gender +
              interviewee_age +
              (1 | province/cfmg_name),
              family = binomial(link = "logit"),
              data = test.data
)

emm.M2 <- emmeans(M2, specs = ~ interviewee_role)
pairwise.M2 <- pairs(emm.M2, adjust = "Tukey")

```

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]

Family: binomial (logit)

Formula:

expand_cfma ~ interviewee_role + interviewee_gender + interviewee_age +
(1 | province/cfmg_name)

Data: test.data

AIC	BIC	logLik	deviance	df.resid
300.1	331.4	-141.1	282.1	229

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.0745	-0.6347	-0.4089	0.7636	2.7635

Random effects:

Groups	Name	Variance	Std.Dev.
cfmg_name:province	(Intercept)	1.0460	1.0227
province	(Intercept)	0.1193	0.3454

Number of obs: 238, groups: cfmg_name:province, 72; province, 8

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.787932	0.977669	-0.806	0.4203
interviewee_rolecfmg_exec_member	-1.294635	0.601186	-2.153	0.0313 *
interviewee_roletrad_leader	-0.547859	0.586662	-0.934	0.3504
interviewee_rolédfo	-1.447932	0.632550	-2.289	0.0221 *
interviewee_roleNGO_rep	-2.052228	0.736019	-2.788	0.0053 **
interviewee_gendermale	0.915355	0.493711	1.854	0.0637 .

```
interviewee_age          0.005378    0.014490    0.371    0.7105
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Correlation of Fixed Effects:

```
      (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -0.395
intrvw_rlt_ -0.271  0.633
intrvw_rldf -0.515  0.625  0.616
intrvw_NGO_ -0.474  0.533  0.479  0.575
intrvw_gndr -0.382 -0.030 -0.073 -0.084   -0.050
interview_g -0.741  0.027 -0.158  0.226    0.247 -0.041
```

contrast	estimate	SE	df	z.ratio	p.value
cfmg_member - cfmg_exec_member	1.295	0.601	Inf	2.153	0.1977
cfmg_member - trad_leader	0.548	0.587	Inf	0.934	0.8838
cfmg_member - dfo	1.448	0.633	Inf	2.289	0.1483
cfmg_member - NGO_rep	2.052	0.736	Inf	2.788	0.0423
cfmg_exec_member - trad_leader	-0.747	0.509	Inf	-1.467	0.5841
cfmg_exec_member - dfo	0.153	0.535	Inf	0.287	0.9985
cfmg_exec_member - NGO_rep	0.758	0.657	Inf	1.153	0.7780
trad_leader - dfo	0.900	0.536	Inf	1.679	0.4471
trad_leader - NGO_rep	1.504	0.687	Inf	2.190	0.1834
dfo - NGO_rep	0.604	0.638	Inf	0.948	0.8781

Results are averaged over the levels of: interviewee_gender

Results are given on the log odds ratio (not the response) scale.

P value adjustment: tukey method for comparing a family of 5 estimates

Defined user groups

Among those CFMGs that had recognised user groups, they were involved in managing the following resources (Figure 8).

```
prefix <- "cfmg_governance_forest_resources_managed_usergroups_"

resource_labels <- c(
  "timber"      = "Timber",
  "poles"       = "Poles",
  "firewood"    = "Firewood",
  "charcoal"    = "Charcoal",
  "honey"       = "Honey",
```



```

"mushrooms"      = "Mushrooms",
"wild_fruits"    = "Wild fruit",
"caterpillars"   = "Caterpillars",
"chikanda"       = "Chikanda",
"munkoyo"        = "Munkoyo",
"lusala_busala"  = "Lusala busala",
"mungongo"       = "Mungongo",
"devils_claw"    = "Devil's claw",
"fish"           = "Fish",
"pupwe"          = "Pupwe",
"brooms"         = "Brooms",
"bamboo"         = "Bamboo",
"medicinal_plants" = "Medicinal plants",
"bushmeat"       = "Wild meat",
"resins_gums"    = "Resins",
"thatching_grass" = "Thatching",
"fibers"         = "Fibres",
"eco_tourism"    = "Eco-tourism"
)

full_var_names <- paste0(prefix, names(resource_labels))

wc_data <- cfm_g |>
  filter(interview_type == "cfm_g_executive") |>
  select(all_of(full_var_names)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = resource_labels[sub(prefix, "", variable, fixed = TRUE)]) |>
  filter(freq > 0)

set.seed(42)
ggplot(wc_data, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 30)) +
  scale_colour_gradient(low = "#6ABFB8", high = "#1A6B63") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))

```



Figure 8: Forest resources managed by user groups within CFMGs, as reported by CFMG executives. Word size is proportional to the number of CFMGs reporting a user-group for that resource type.

Evidently, people are harvesting a wide variety of products from CFs, although honey and mushrooms appear especially important. It is not clear from the responses, if these user-groups are defined within the CF management plan or whether these are informal groups that are nevertheless recognised by the CFMG.

Governance index

To summarise CFMG governance performance, we constructed a governance index (0–10) based on the presence or absence of ten features reported by CFMG executives: receipt book, bank account, financial reporting, annual reporting, issuing permits, HFOs appointed, work-plan, budget, business plan, and term limits. We then tested whether two aspects of external support — the frequency of DFO visits and the presence of a partner organisation — were associated with higher governance scores (Figure 9).

```
load("cfmg.RData")

cfmg |>
  filter(interview_type == "cfmg_executive") |>
  distinct(cfmg_name, .keep_all = TRUE) |>
  drop_na(gov_index) |>
  ggplot(aes(x = gov_index)) +
  geom_histogram(binwidth = 1, fill = "#2A9D8F", colour = "white", linewidth = 0.4) +
  scale_x_continuous(breaks = 0:10, name = "Governance index (out of 10)") +
  scale_y_continuous(name = "Number of CFMGs") +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5)
  )
```

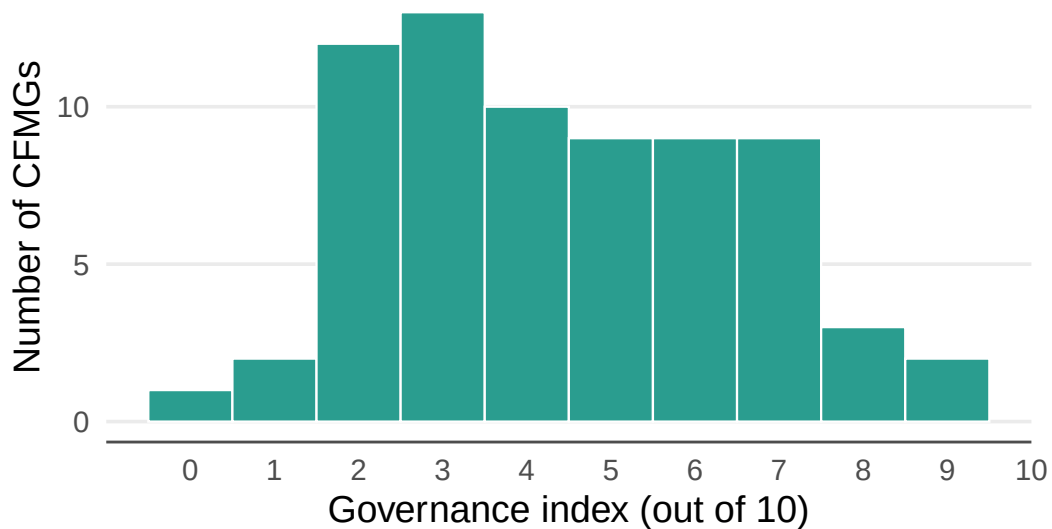


Figure 9: Distribution of CFMG governance index scores (0–10). Each point represents one CFMG. The index counts the number of ten governance features reported as present by the CFMG executive FGD.

Table 6: Linear mixed model of the CFMG governance index. Predictor coefficients, standard errors and p-values are shown. Random intercept variance for province is reported at the bottom.

Table 7: Predictors of CFMG governance index

	(1)
Intercept	2.418 (0.693)***
DFO visit frequency (1–5)	0.909 (0.216)***
External partner (yes)	0.054 (0.542)
SD (Intercept province)	0.774
SD (Observations)	1.784
Num.Obs.	67

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Standard errors in parentheses.

i Governance index model — methods

We modelled the governance index as a continuous response using a linear mixed model (LMM) with random intercepts for province. Because the governance index is measured at the CFMG level (one value per CFMG), nesting CFMGs within province as a second random effect is not possible; province is the only grouping variable. DFO visit frequency was treated as a numeric predictor (1 = never, 2 = rarely, 3 = occasionally, 4 = often, 5 = very often; don't know excluded). Presence of an external partner organisation was treated as a binary predictor (yes = 1, no = 0). We used `lmer` (lme4) and examined coefficient significance with `lmerTest`.

```
M_gov <- lmer(
  gov_index ~ dfo_visits_num + external_partner + (1 | province),
  data = exec_cfmg,
  REML = TRUE
)
```

```
p_dfo <- exec_cfmg |>
  drop_na(dfo_visits_num, gov_index) |>
  mutate(dfo_label = case_match(
    dfo_visits_num,
    1 ~ "Never", 2 ~ "Rarely", 3 ~ "Occasionally", 4 ~ "Often", 5 ~ "Very often"
  ),
  dfo_label = factor(dfo_label,
```

```

    levels = c("Never", "Rarely", "Occasionally", "Often", "Very often")))) |>
  ggplot(aes(x = dfo_label, y = gov_index)) +
  geom_boxplot(fill = "#76B7A0", colour = "grey30", outlier.shape = NA, width = 0.5) +
  geom_jitter(width = 0.15, height = 0.2, colour = "#1A6B63", alpha = 0.6, size = 1.8) +
  scale_y_continuous(name = "Governance index", limits = c(0, 10), breaks = seq(0, 10, 2)) +
  scale_x_discrete(name = "DFO visit frequency") +
  labs(tag = "(a)") +
  theme_minimal(base_size = 13) +
  theme(
    panel.grid.major.x = element_blank(),
    panel.grid.minor    = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5)
  )

p_partner <- exec_cfm |>
  drop_na(external_partner, gov_index) |>
  mutate(partner_label = if_else(external_partner == 1, "Yes", "No")) |>
  ggplot(aes(x = partner_label, y = gov_index)) +
  geom_boxplot(fill = "#76B7A0", colour = "grey30", outlier.shape = NA, width = 0.4) +
  geom_jitter(width = 0.12, height = 0.2, colour = "#1A6B63", alpha = 0.6, size = 1.8) +
  scale_y_continuous(name = NULL, limits = c(0, 10), breaks = seq(0, 10, 2)) +
  scale_x_discrete(name = "External partner organisation") +
  labs(tag = "(b)") +
  theme_minimal(base_size = 13) +
  theme(
    panel.grid.major.x = element_blank(),
    panel.grid.minor    = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5)
  )

gridExtra::grid.arrange(p_dfo, p_partner, ncol = 2, widths = c(2, 1))

```

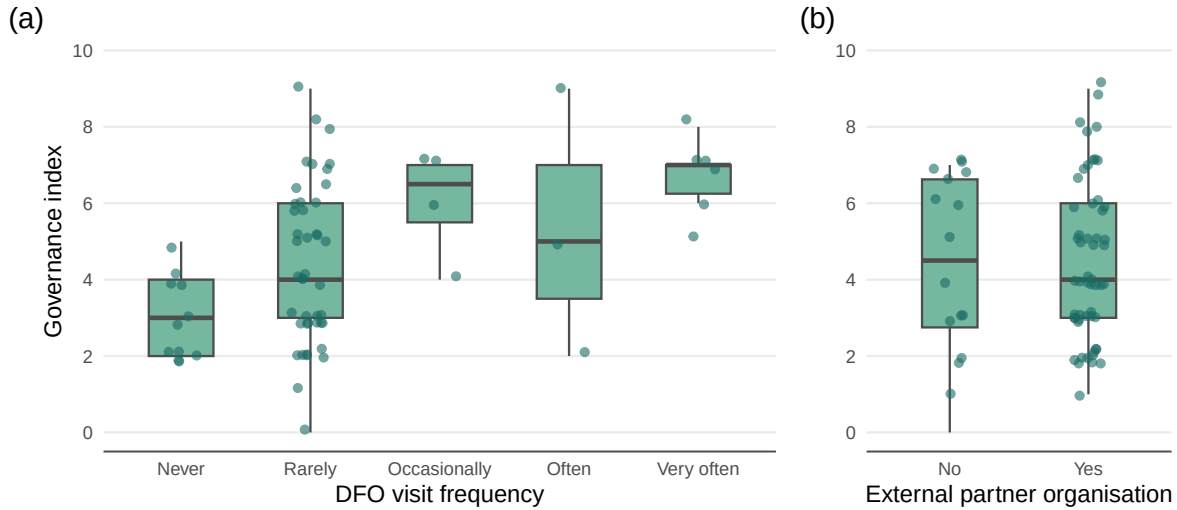


Figure 10: CFMG governance index in relation to (a) DFO visit frequency and (b) presence of an external partner organisation. Points are individual CFMGs, jittered slightly. Boxes show median and interquartile range.

Discussion

In this chapter we assessed CFMG governance across five domains: legal processes, governance processes, business processes, external support, and community buy-in as reflected by interest in expanding the CF and the activities of user-groups.

The most consistent pattern across all three process domains — legal, governance, and business — is that foundational steps have been taken by most CFMGs, but the more demanding, ongoing requirements are often incomplete. Early legal steps, such as obtaining a recognition letter and signing a forest management agreement, have been completed by the large majority of CFMGs, but later and more involved documents — the registration certificate, forest-use permit, and benefit-sharing agreement — are absent in a substantial proportion. Similarly with respect to governance processes, most CFMGs have held elections and have term limits in place, but basic management actions — opening bank accounts, operating receipt books, and financial and annual reporting — are frequently missing. Business processes are the weakest area of all. Annual workplans, budgets and business plans are rarely in place. Without these planning and financial management tools, CFMGs cannot sustainably manage their resources, are poorly positioned to account for and redistribute income — including carbon payments — and will struggle to demonstrate compliance with the forest management license or to agreements with external partners.

The fact that most CFMGs are overdue for elections is particularly concerning. Timely elections are a basic requirement of the CFMG Bylaws & Constitution, and overdue elections

undermine the democratic legitimacy of the CFMG executives and could contribute to the perception of elite capture noted elsewhere in this report. These perceptions may be very valid in some cases. The enumerators noted that while some CFMGs had not held elections because they were no longer active, the CFMG executives of others stated that the CFMG had yet to receive carbon income and they were unwilling to hold elections, because they anticipated personal benefits from being a CFMG executive member once the carbon income started to flow. Still others had received carbon income and were unwilling to give up their positions because they do receive some kind of personal benefit. It is essential that this kind of corruption is stamped out, as it may threaten the legitimacy of CFM in the eyes of the community and potentially undo gains made in SFM and sustainable development. Ensuring that CFMGs run elections on schedule and fairly should be a priority for the FD and for supporting partners.

External support from the FD is insufficient. Most CFMGs are rarely or never visited by DFOs or other district forestry staff, despite the statutory duty of the FD to implement the CFM policy. This gap is partly compensated by conservation NGOs and carbon trading enterprises, many of which visit their partner CFMGs more frequently. However, this support is not universal: a significant proportion of CFMGs in our sample had no external partner. Given the governance deficits identified above, regular external support from CFMG partners for capacity building and oversight is clearly a necessity. The current model, in which CFM is effectively co-governed by the community and the private sector rather than the state, may be adequate for CFMGs with active and capable partners but leaves others without meaningful support.

Interest in expanding the CF is highest among CFMG executives and drops markedly among community members, particularly youth. Among individual interviewees, NGO representatives are significantly more enthusiastic about expansion than traditional leaders or community members. This divergence — consistent with the equity chapter findings — suggests that the CFMG leadership and external partner stakeholders have a more optimistic view of CFM outcomes than the broader community. Although possibly not widespread at the moment, poor governance could lead to increased levels of community dissatisfaction with CFM and, if not addressed, could pose a long-term risk to the legitimacy and sustainability of CFM.

In conclusion, CFMG governance is still maturing. The legal framework for CFM is largely in place, but institutional and business management processes lag behind. Strengthening financial management, ensuring timely elections, expanding DFO and district forestry staff engagement, and closing the gap between executive and community perspectives on CFM will be essential for CFM to fulfil its potential in Zambia.